Assignment #4
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Summarized Report:

Trying to minimize the **Tour Length** in a Traveling Salesman Problem (TSP) using Ant Colony Optimization.

Run Summary

- Best Tour Found by City Indexes:
  28 -> 22 -> 20 -> 21 -> 17 -> 18 -> 14 -> 11 -> 10 -> 9 -> 5 -> 1 -> 0 -> 4 -> 7 -> 3 -> 2 -> 6 -> 8 -> 12 -> 13 -> 16 -> 27 -> 25 -> 19 -> 15 -> 24 -> 26 -> 23 -> 28
- Best Tour Length: **4.05511616589579**

Convergence Summary:

Iteration	Best Tour Length
1	**4.506**
2	**4.257**
5	**4.156**
7	**4.109**
13	**4.076**
21	**4.055**
50	**4.055 *(final)***

Parameters Used:

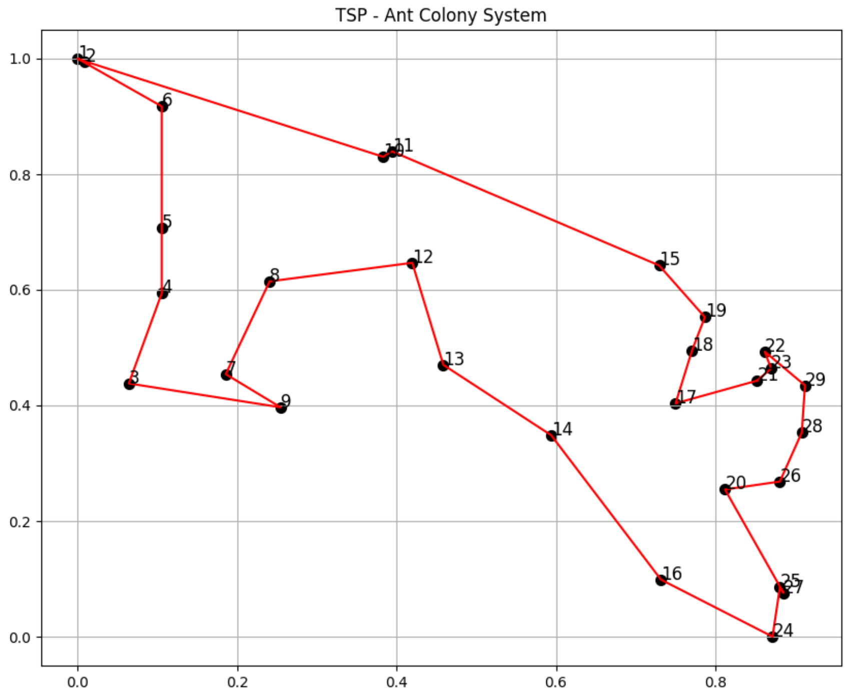
Parameter	Value
Alpha (α) - pheromone influence	1
Beta (β) - heuristic influence	5
Rho (ρ) - evaporation rate	0.5
Number of Ants (m)	10
Iterations	50

Tools and Technologies:

- Python
- Numpy Library
- Random Library
- Matplotlib (for visualizations).

I have put a random seed to ensure the reproducibility of the code per each run:
np.random.seed(X)

The final Plotting:



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